

Engineering report template

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Description

This document itends to document blah blah blah

Standard packages

```
using Pkg
# Ensure packages are available in the current environment
# Pkg.add(["DataFrames", "CSV", "Plots", "ControlSystems", "LinearAlgebra", "LaTeXStrings"])

using LinearAlgebra, Statistics
using DataFrames, CSV
using Plots
plotly()
using PlotThemes
# Detect if we are rendering in a headless/build environment like Quarto
const IS_RENDERING = haskey(ENV, "QUARTO_ROUTES") || !isinteractive()
if !IS_RENDERING
    # --- Light/High-Contrast Settings for Quarto HTML ---
    theme(:default) # Swaps to a clean, light-friendly palette
    default(
        background_color          = :white,
        background_color_inside   = :white,
        background_color_legend   = RGBA(0, 0, 0, 0.1), # Keep a slight dark tint for legend
                                readability
        foreground_color_legend    = nothing,

        # --- Grid & Frame Tuning ---
        grid                      = true,
        gridalpha                  = 0.12,                # Keep grid lines faint so they sit nicely over your
                                theme
        linewidth                  = 1.3,

        # --- Layout & Geometry ---
        size                      = (600, 300),          # Good aspect ratio for a horizontal 1x2 layout
        legend                    = :best
    )
else
    theme(:dark)
    default(
        # --- The Transparency Magic ---
        background_color          = :transparent, # Makes the main plot area transparent
        background_color_inside   = :transparent, # Makes the inside of the axes transparent
        background_color_legend   = RGBA(0, 0, 0, 0.4), # Keep a slight dark tint for legend
                                readability
        foreground_color_legend    = nothing,
```

```

# --- Grid & Frame Tuning ---
grid      = true,
gridalpha = 0.12,      # Keep grid lines faint so they sit nicely over your
                        theme
linewidth = 1.3,

# --- Layout & Geometry ---
size      = (600, 300), # Good aspect ratio for a horizontal 1x2 layout
legend    = :best
)
end

using LaTeXStrings # Essential for math in labels

println("Julia Engineering Stack Loaded.")

```

Julia Engineering Stack Loaded.

Topic 1

Insert text for equation here. You can number and reference equation with [Equation 1](#)

$$x = a_2 \tag{1}$$

Sample plot code below

```

x = 0:0.1:10
y = sin.(2*x)

# Create a line plot
plot(x, y, title="Sine Wave", xlabel="Time", ylabel="Amplitude", label="sin(x)", sp=1,
      alpha=0.5)

```

